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Connector set mating method for connecting two plug connectors to two socket connectors and a connector set using the same

Abstract

Each of two plug connectors includes a main body configured to retain a plurality of electric wires and a pair of push members protruding from both sides of the main body along a first direction. In connector mating operation, the two plug connectors are respectively inserted into associated receptacle connectors in a state where the two plug connectors are mechanically coupled or bound to a relative positional relationship by a jig in a position where the pair of push members in one of the two plug connectors are displaced from the pair of push members in the other of the two plug connectors along a second direction orthogonal to the first direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4998884	12/1990	Ishikawa	439/372	H01R 13/629
6176745	12/2000	Furutani	439/701	H01R 13/506
2009/0042445	12/2008	Ichio et al.	N/A	N/A
2017/0346230	12/2016	Imai	N/A	H01R 13/639
2019/0103705	12/2018	Amemiya et al.	N/A	N/A
2020/0014146	12/2019	Aoshima et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1-113976	12/1988	JP	N/A
7-42016	12/1994	JP	N/A
2009-43642	12/2008	JP	N/A
2016-103393	12/2015	JP	N/A
2016-122563	12/2015	JP	N/A
2019-67744	12/2018	JP	N/A
2019-87408	12/2018	JP	N/A
2020-9614	12/2019	JP	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2021-046094 filed on Mar. 19, 2021, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

(2) The present specification discloses a connector mating method for respectively inserting and mating a first plug connector into a first receptacle connector and a second plug connector into a second receptable connector, and also discloses a connector set comprising the first plug connector and the second plug connector.

BACKGROUND

(3) Typically, connector devices include a receptacle connector having a recess for connection and a plug connector having a protrusion to be inserted into the recess of the receptacle connector. Some plug connectors retain a plurality of electric wires aligned along a predetermined arrangement direction.

(4) For example, JP 2019-87408 A (Patent Document 1) discloses a plug connector (which is referred to as a “first connector” in Patent Document 1) equipped with a housing for retaining a plurality of electric wires which are aligned along a predetermined arrangement direction. It has been desired that the size of such plug connectors be further minimized. In particular, when a plug connector is utilized in a situation where a great number of electronic devices are arranged in a limited space, as in the case of a vehicle, it is desirable to further minimize the plug connector.

CITATION LIST

Patent Literature

(5) Patent Document 1: JP 2019-87408 A

(6) Under the circumstances, attempts to decrease dimensions of plug connectors have been made conventionally. For example, in the plug connector of Patent Document 1, its dimensions along a thickness direction orthogonal to the arrangement direction are reduced as much as possible. In the above-described configuration, however, a region which is pushed by fingers in order to mate the plug connector with the receptacle connector is limited to only two points located on both sides of a plurality of electric wires in the arrangement direction. When only the two points are pushed as described above, the plug connector is easily inclined about a line passing through the two points. As a result, it has been difficult to appropriately insert the plug connector into the receptable connector.

(7) In other words, minimization of the plug connector conventionally results in poor workability of operation to insert the plug connector into the receptacle connector.

(8) In view of the circumstances, the present specification discloses a connector mating method and a connector set with which workability in operation of inserting a plug connector into a receptacle connector can be improved while maintaining the plug connector in a thin shape.

SUMMARY

(9) A connector mating method disclosed in this specification is a method for inserting a first plug connector into a first receptacle connector and a second plug connector into a second receptacle connector to mate the connectors, in which each of the first plug connector and the second plug connector includes a main body configured to retain a plurality of electric wires and a pair of push members protruding from both sides of the main body along a first direction, and in which the first plug connector and the second plug connector are mechanically coupled to each other or bound by a jig to a relative positional relationship therebetween in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug

connector along a second direction orthogonal to the first direction, and are inserted into the first plug connector and the second plug connector while being mechanically coupled to each other or being bound to the relative positional relationship by the jig.

(10) In the above-described configuration, a total of four push members disposed on the first and second plug connectors are pushed while maintaining a positionally restricted relationship therebetween. This makes the first and second plug connectors less prone to inclination. As a result, workability in operation to insert the plug connector into the receptacle connector can be improved while maintaining a small size of each of the plug connectors.

(11) In this case, a plurality of wire connection holes through which the electric wires are inserted may be arranged in a line along the first direction.

(12) When the plurality of wire connection holes are arranged in the line along the same direction as an arrangement direction in which the pair of push members are arranged (i.e., the first direction), the plug connector can be configured to have smaller dimensions in the second direction, and accordingly have a thin outline.

(13) In an aspect of this disclosure, the first plug connector and the second plug connector may be mechanically coupled to each other or bound by the jig to the relative positional relationship in a position where the first plug connector and the second plug connector are adjacent to each other in the second direction.

(14) When configured as described above, the first and second plug connectors mated with the first and second receptacle connectors can be made compact in size.

(15) In an aspect of the present disclosure, each of the first plug connector and the second plug connector may further include a guide rib extending along an insertion direction of the plug connectors, the guide rib being designed to be received in a guide groove defined in an associated one of the receptacle connectors. Further, a relative position between the guide rib and the main body in the first plug connector may be varied from a relative position between the guide rib and the main body in the second plug connector, and when one of the plug connectors tries to enter an unassociated one of the receptacle connectors, the guide rib interferes with a part of the unassociated one of the receptacle connectors to thereby block further entry of the plug connector.

(16) The above-described configuration can reliably prevent an insertion error where the plug connector is inserted into an unassociated one of the receptacle connectors.

(17) A connector set disclosed herein includes a first plug connector configured to be inserted into a first receptacle connector and a second plug connector configured to be inserted into a second receptacle connector, in which each of the first plug connector and the second plug connector has a main body configured to retain a plurality of electric wires and a pair of push members protruding along a first direction from both sides of the main body. Further, in the connector set, the first plug connector is mechanically connectable to the second plug connector in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction.

(18) When configured as described above, a total of four push members disposed on the first and second plug connectors are pushed while maintaining the positionally restricted relationship therebetween. This makes the first and second plug connectors less prone to inclination about a line. As a result, workability in operation to insert the plug connectors into the receptacle connectors can be improved while maintaining a small size of each of the plug connectors.

(19) Another connector set disclosed herein includes a first plug connector configured to be inserted into a first receptacle connector, a second plug connector configured to be inserted into a second receptacle connector, and a jig, in which each of the first plug connector and the second plug connector includes a main body configured to retain a plurality of electric wires and a pair of push members protruding along a first direction from both sides of the main body. Further, in the other connector set, the jig includes a binding member which is configured to house a part of each of the first and second plug connectors in such a manner that the first and second plug connectors

are bound to a relative positional relationship therebetween in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction.

(20) When configured as described above, a total of four push members disposed on the first and second plug connectors are pushed while maintaining the positionally restrained relationship therebetween. This makes the first and second plug connectors less prone to inclination about a line. In this way, workability in operation to insert the plug connectors into the receptacle connectors can be improved while maintaining a small size of each of the plug connectors.

(21) A technique disclosed herein can ensure that the workability in operation to insert the plug connectors into the receptacle connectors is further improved while maintaining the small size of each of the plug connectors.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Embodiments of the present disclosure will be described based on the following figures, wherein:

(2) FIG. 1 is a perspective view of a connector set;

(3) FIG. 2 is a perspective view of a terminal base;

(4) FIG. 3 is a perspective view of a first plug connector;

(5) FIG. 4 is a perspective view of a second plug connector;

(6) FIG. 5 is a perspective view of another connector set;

(7) FIG. 6 is a perspective view of another terminal base;

(8) FIG. 7 is a perspective view of a first plug connector in the connector set shown in FIG. 5;

(9) FIG. 8 is a perspective view of a second plug connector in the connector set shown in FIG. 5; and

(10) FIG. 9 is a perspective view of a jig which is flipped upside down from a state shown in FIG. 5.

DESCRIPTION OF EMBODIMENTS

(11) Hereinafter, a connector set **10** incorporating two plug connectors composed of a first plug connector **12f** and a second plug connector **12s**, and a connector mating method using the connector set **10** will be described with reference to the drawings. FIG. 1 shows a perspective view of the connector set **10**, and FIG. 2 shows a perspective view of a terminal base **15** which has two receptacle connectors **14f** and **14s** to receive the connector set **10**. FIGS. 3 and 4 respectively show perspective views of the first plug connector **12f** and the second plug connector **12s**. In the following description, a direction in which the plug connectors **12f** and **12s** are inserted into the receptacle connectors **14f** and **14s** is referred to as an “insertion direction”, a direction in which a pair of push members **22**, which will be described below, are arranged is referred to as a “first direction”, and a direction orthogonal to both the insertion direction and the first direction is referred to as a “second direction”. Further, in the following description, a posterior part of the insertion direction is considered a “downside”, and an anterior part of the insertion direction is considered an “upside”. However, the terms “downside” and “upside” are irrelevant to gravitational downside and upside. In the drawings, the “first direction”, the “second direction”, and the “insertion direction” are described as “Y”, “X”, and “Z”, respectively.

(12) The connector set **10** includes two plug connectors consisting of the first plug connector **12f** and the second plug connector **12s**. The terminal base **15** has a first receptacle connector **14f** in which the first plug connector **12** is inserted and a second receptacle connector **14s** in which the second plug connector **12s** is inserted. In this example, the two receptacle connectors **14f** and **14s** are merged with each other to form one large recess. Each of the plug connectors **12f** and **12s** is

inserted into an associated one of the receptacle connectors **14f** and **14s** and fitted therein to electrically connect on-board electronic devices installed in a vehicle to other on-board electronic devices.

(13) More specifically, the first plug connector **12f** is connected to a first temperature sensor (not illustrated) via four electric wires **18f** (in FIG. 1, only one of the electric wires **18f** is illustrated, and is omitted in FIG. 3). The second plug connector **12s** is connected to a second temperature sensor (not illustrated) via four electric wires **18s** (in FIG. 1, only one of the electric wires **18s** is illustrated, and is omitted in FIG. 3). The first temperature sensor is installed in a not-illustrated first rotating electric machine which is mounted on the vehicle to detect the temperature of the first rotating electric machine. Similarly, the second temperature sensor is incorporated in a not-illustrated second rotating electric machine which is mounted on the vehicle to detect the temperature of the second rotating electric machine. When the plug connectors **12f** and **12s** are inserted into the receptacle connectors **14f** and **14s**, the first and second temperature sensors are electrically connected to other electronic devices and become able to transmit and receive electric power and electric signals.

(14) The first plug connector **12f** and the second plug connector **12s**, which are completely separated independent parts, are mechanically coupled and merged into one unit as shown in FIG. 1 when the plug connectors **12f** and **12s** are inserted into the corresponding receptacle connectors **14**, respectively. Configurations of the first plug connector **12f** and the second plug connector **12s** are explained in detail below.

(15) The first plug connector **12f** has a resin housing **16f** and an electric mechanism (not illustrated) arranged within the resin housing **16f**. The electric mechanism includes components, such as an input-side contact to be brought into contact with the electric wire **18f**, an output-side contact contactable with a contact of the first receptacle connector **14f**, and a connection line for electrically connecting the input-side contact and the output-side contact. As the components can be implemented by conventionally known techniques, details of the components are not explained herein.

(16) The housing **16f** is roughly divided into five portions consisting of a main body **20f**, a push member **22f**, a retainer lug **24**, a guide rib **30**, and a coupler member **32**. The main body **20f** is an approximately rectangular member configured to retain the electric mechanism and a plurality of electric wires **18f**. An upper end surface of the main body **20f** located on an upper side in the insertion direction (Z direction) is provided with four wire connection holes **26f** which are arranged in a row along the first direction (Y direction). The electric wires **18f** are respectively inserted into the wire connection hole **26f**. The electric wires **18f** inserted into the wire connection holes **26f** are secured in a state of being in contact with the input-side contact. Dimensions of the main body **20f** in the second direction (X direction) and the first direction are defined as small as possible within a range capable of maintaining appropriate strength. As a result, the upper end surface of the main body **20f** is in most areas occupied by the wire connection holes **26f**, leaving almost no area for fingers to push the main body **20f** on the upper end surface.

(17) The retainer lug **24** is fixed to an end surface of the main body **20** that is located at one of second direction ends of the main body **20**. The retainer lug **24** is a cantilever arm type member which is fixed only at its lower portion to the main body **20f**, and is configured to be swingable through flexure in the second direction. When the first plug connector **12f** is inserted into the first receptacle connector **14f**, a part of the retainer lug **24** is engaged with a part of the first receptacle connector **14f**. This can prevent unintended removal of the first plug connector **12f** from the first receptacle connector **14f**. The first plug connector **12f** can be withdrawn and removed from the first receptacle connector **14f** by pulling the first plug connector **12f** upward in a state where the retainer lug **24** is swung toward the main body **20f**.

(18) A pair of push members **22f** are protruded outward along the first direction from side surfaces of the main body **20f** located at both first direction ends thereof. Upper end surfaces of the push

members **22f** function as pushing surfaces **28f** which are pushed by fingers in order to insert the first plug connector **12f** into the first receptacle connector **14f**. The pushing surfaces **28f** are flush with the upper end surface of the main body **20f**, i.e. the surface in which the wire connection holes **26f** are defined.

(19) The coupler member **32** is designed to be engaged with a part of the second plug connector **12s** for coupling the first plug connector **12f** to the second plug connector **12s**. The coupler member **32** includes a coupling loop **34** (see FIG. 3) and a plurality of slide claws **36**. It should be noted that, in FIG. 3, the coupling loop **34** is located at an invisible position, and is therefore indicated by a broken line in the drawing.

(20) The coupling loop **34** is outwardly protruded along the second direction from an end surface of the main body **20f** that is located at the other of the second direction ends of the main body **20f** (i.e., the end surface opposite to the retainer lug **24**), and is configured to define a vertical through hole between the end surface of the main body **20f** and the coupling loop **34** itself. A coupling lug **42** in the second plug connector **12s**, which will be described below, enters the hole to engage with the coupling loop **34**.

(21) The slide claws **36** are outwardly protruded along the second direction, at both first direction ends, from the end surface of the main body **20f** that is located at the other of the second direction ends of the main body **20f**. A total of four slide claws **36** are arranged in a two-by-two matrix in which the slide claws **36** are spaced from each other in both the first direction and the vertical direction. Each of the slide claws **36** is formed in an approximately L-letter shape with its tip end being bent inward in the first direction. When such tip ends of the slide claws **36** are fitted in below-described slide rails **44** of the second plug connector **12s**, the first plug connector **12f** is allowed to move relative to the second plug connector **12s** along the insertion direction, while being restricted so as not to move along the first direction or the second direction.

(22) Guide ribs **30f** extending along the insertion direction are formed on both end surfaces of the main body **20f** that are located at first direction ends of the main body **20f**. In this example, two guide ribs **30f** are disposed for each of the end surfaces, and a total of four guide ribs **30** are provided to the entire first plug connector **12f**. The first receptacle connector **14f** has guide grooves **52f** which are configured to receive the guide ribs **30f**. When the guide ribs **30f** are inserted into the guide grooves **52f**, movement of the first plug connector **12f** is guided in the vertical direction (i.e., the insertion direction).

(23) Next a configuration of the second plug connector **12s** will be described. The second plug connector **12s** includes, as in the case of the first plug connector **12f**, a resin housing **16s** and a not-illustrated electric mechanism installed within the housing **16s**. The housing **16f** includes a main body **20s**, push members **22s**, a coupling member **38**, and guide ribs **30s**. The main body **20s** and the push members **22s** are almost identical in structure to the main body **20f** and the push members **22f** of the first plug connector **12f**, and explanation related to these components is not repeated.

(24) The coupling member **38** includes a pair of rail members **40** and the coupling lug **42**. The rail members **40** are formed in a flat plate shape extending outward in the second direction from an end surface of the main body **20s** that is located on one of second direction ends of the main body **20s**. The pair of rail members **40** are spaced from each other in the first direction. A slide rail **44** extending along the vertical direction is formed on an outer end surface facing outward in the first direction for each of the rail members **40**. The slide rail **44** extends, as shown in FIG. 4, penetrating through an upper end surface of the rail member **40**. Therefore, the tip end of the slide claw **36** in the first plug connector **12f** can be inserted from above the second plug connector **12s** into the slide rail **44**. On the other hand, the slide rail **44** does not reach a lower end of the rail member **40**, and a stopper surface **45**, which is a flat plane facing upward, is defined in the rail member **40** at a height that matches a lower end of the slide rail **44**. The stopper surface **45** blocks the slide claw **36** to prevent the first plug connector **12f** from sliding further below the stopper surface **45**.

(25) The coupling lug **42** is disposed between the pair of rail members **40**. The coupling lug **42** is a

lug member protruding upward. The coupling lug **42** has a hitching portion **42a** protruding along the second direction in the vicinity of a tip end of the coupling lug **42**, and an upper area of the coupling lug **42** located above the hitching portion **42a** is formed in a tapered shape whose dimension in the second direction is gradually decreased toward the top of the coupling lug **42**.

(26) Similarly to the guide ribs **30f**, the guide ribs **30s** extending along the insertion direction are formed on both end surfaces of the main body **20s** that are located on first direction ends of the main body **20s**. Guide grooves **52s** for receiving the guide ribs **30s** are defined in the second receptacle connector **14s**.

(27) When the first plug connector **12f** and the second plug connector **12s** are inserted into the first receptacle connector **14f** and the second receptacle connector **14s**, the first plug connector **12f** and the second plug connector **12s** are coupled and merged into one unit. Specifically, in operation to couple the plug connectors **12f** and **12s**, the slide claws **35** are fitted into the slide rails **44** from above the side rails **44**, and the first plug connector **12f** is slid downward on the second plug connector **12s**. The downward sliding of the first plug connector **12f** is guided by the slide rails **44**. The coupling lug **42** enters the hole in the coupling loop **34** at some point during the sliding of the first plug connector **12f**.

(28) Then, the first plug connector **12f** is slid further downward until the slide claws **36** located on a lower part make contact with the stopper surface **45**. At this time, the hitching portion **42a** passes through the coupling loop **34** to surmount the coupling loop **34**. The contact of the lower slide claws **36** with the stopper surface **45** hampers further downward movement of the first plug connector **12f**. Meanwhile, when the hitching portion **42a** has surmounted the coupling loop **34**, the hitching portion **42a** and the coupling loop **34** are engaged with each other, to thereby hamper upward movement of the first plug connector **12f**. In other words, when the lower slide claws **34** are brought into contact with the stopper surface **45**, the first plug connector **12f** becomes unable to move relative to the second plug connector **12s**, and the first and second plug connectors **12f** and **12s** are mechanically coupled to each other accordingly.

(29) Dimensions of each section in the coupler member **32** and the coupling member **34** are defined in such a manner that the two plug connectors **12f** and **12s** are aligned at the same height when coupling between the plug connectors **12f** and **12s** is complete. Therefore, in this example, a total of four pushing surfaces **28f**, **28s** are located on the same plane when the two plug connectors **12f** and **12s** are completely coupled to each other. In addition, when the coupling is complete, the pair of pushing surfaces **28f** in the first plug connector **12f** are displaced in the second direction from the pair of pushing surfaces **28s** in the second plug connector **12s**.

(30) After coupling the two plug connectors **12f** and **12s**, an operator inserts the two coupled plug connectors **12f** and **12s** into the hole in the receptacle connectors **12f** and **14s**. In this inserting operation, the operator pushes the four pushing surfaces **28f** and **28s** from thereabove with their hands. Then, mating of the connectors is finally complete when the plug connectors **12f** and **12s** are inserted into the receptacle connectors **14f** and **14s** to a position where the retainer lug **24** is engaged with a part of the first receptacle connector **14f**.

(31) As can be clearly understood from the above description, the two plug connectors **12f** and **12s** are mechanically coupled to each other in order to insert the two plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s** in this example. The reason for mechanically coupling the two plug connectors **12f** and **12s** is described below.

(32) As described above, the first plug connector **12f** in this example is connected to the first temperature sensor, and the second plug connector **12s** is connected to the second temperature sensor. Further, the first temperature sensor is incorporated in the first rotating electric machine, while the second temperature sensor is incorporated in the second rotating electric machine. Both the first rotating electric machine and the second rotating electric machine are mounted on the vehicle. Here, in addition to the rotating electric machines, a variety of electronic devices are mounted on the vehicle, while a space available for mounting such components within the vehicle

is limited. For this reason, it is desired that on-board components including the plug connectors **12f** and **12s** be minimized to the greatest possible extent.

(33) To satisfy such a desire, the main bodies **20f** and **20s** of the plug connectors **12f** and **12s** are reduced in size to a maximum possible extent at which the strength of the main bodies **20f** and **20s** can be properly maintained. As a result, the upper end surfaces of the main bodies **20f** and **20s** are mostly occupied by the wire connection holes **26**, leaving almost no areas available for pushing by fingers. Therefore, in this example, the push members **22** are provided at both first direction ends of the main bodies **20f** and **20s** to secure the pushing surfaces **28f** and **28s** which can be pushed by the fingers.

(34) Here, it is assumed that the first plug connector **12f** is solely inserted into the first receptacle connector **14f** without being mechanically coupled to the first plug connector **12s**. In this case, the operator pushes the two pushing surfaces **28f** located on both sides of the main body **20f** by hand. In other words, only two points on a line along the first direction are pushed by the operator. When pushed in this way, the first plug connector **12f** is apt to sway about the line passing through the two points. As a result, there is a risk that the first plug connector **12f** may be inserted in a state of being inclined relative to the first receptacle connector **14f**. When the first plug connector **12f** is inclined, the guide ribs **30f** butt against the guide grooves **52f**, which hinders further insertion of the first plug connector **12f**. Therefore, it has been necessary for the operator to slightly adjust the first plug connector **12f** to correct for the position every time the first plug connector **12f** is inclined. That is, in a case where the first plug connector **12f** is solely inserted into the first receptacle connector **14f**, because the operator is repeatedly required to adjust the position of the first plug connector **12f** which is apt to be inclined, connector mating work can often be time consuming.

(35) With this in mind, in connector mating operation according to this example, the first plug connector **12f** and the second plug connector **12s** are mechanically coupled to be merged into one unit. As a result, the merged plug connectors **12f** and **12s** are equipped with the four pushing surfaces **28f** and **28s** as a whole. The four pushing surfaces **28f** and **28s** are spaced from each other in both the first direction and the second direction, and are not aligned in a row. In an action of pushing the four pushing points, inclination about a line that passes through two pushing points among the four pushing points can be prevented by a force applied to the other two pushing points. For example, in FIG. 1, if the first plug connector **12f** sways around a line L1 passing through the two pushing surfaces **28f**, the pushing surfaces **28** attempt to move upward. However, when the pushing surfaces **28s** are pushed simultaneously with the pushing surfaces **28f**, the first plug connector **12f** is not caused to sway. Similarly, in a case where the first plug connector **12f** attempts to sway around a line L2 passing through the pushing surface **28f** and the pushing surface **28s** which are located on the right side in FIG. 1, the attempt to sway can be prevented by a downward force applied to the pushing surface **28f** and the pushing surface **28s** which are located on the left side in FIG. 1. This can eliminate the necessity for the operator to repeatedly adjust the position of the plug connectors **12f** and **12s** during operation to insert the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s**, and can thus improve workability of mating the connectors.

(36) Meanwhile, for securing four push points, it is also conceivable that the first plug connector **12f** and the second plug connector **12s** are initially formed as one integral connector rather than being formed as mechanically couplable connectors. That is, it may be considered to prepare a single plug connector **12** including a single main body **20** in which eight wire connection holes **26** are defined, four push members **22** protruded from the single main body **20**, and a single retainer lug **24** fixed to the single main body **20**. Even with this configuration of the plug connector **12**, the four pushing surfaces **28** can be secured for pushing the plug connector **12**, to thereby prevent inclination thereof.

(37) However, in this case, workability in assembling the temperature sensor or the rotating electric machine will be deteriorated. Specifically, the first and second temperature sensors are respectively connected to the first and second plug connectors **12f** and **12s**, and respectively attached, in the

connected state, to the first and second rotating electric machines. Further, the first and second rotating electric machines incorporating the first and second temperature sensors are installed in the vehicle. In a case where the first plug connector **12f** and the second plug connector **12s** are integrated into one plug connector **12**, the first temperature sensor has been mechanically coupled to the second temperature sensor via the plug connector **12** and the electric wires **18**. In this case, an action of moving one of the first and second temperature sensors is transferred to the other of the first and second temperature sensors, which often causes unintentional behavior of the temperature sensors. As a result, the operator is not able to move the temperature sensors and the rotating electric machines as desired, resulting in deteriorated workability in installation.

(38) With this in mind, in this example, the first plug connector **12f** and the second plug connector **12s** are formed as completely separated independent components, and mechanically coupled to each other only when the connectors are mated. In this way, it becomes possible to improve workability in mating the connectors while improving workability in installing the temperature sensors and the rotating electric machines.

(39) Next, another connector set **10** and another connector mating method are explained with reference to FIGS. **5** to **9**. FIG. **5** shows a perspective view of the connector set **10**, and FIG. **6** shows a perspective view of a terminal base **15**. FIG. **7** shows a perspective view of a first plug connector **12f**, and FIG. **8** shows a perspective view of a second plug connector **12s**. In addition, FIG. **9** shows a perspective view of a jig **60** which is flipped upside down from a state shown in FIG. **5**.

(40) The connector set **10** shown in FIG. **5** includes the jig **60** in addition to the first plug connector **12f** and the second plug connector **12s**. The jig **60** is designed to bind a relative positional relationship between the first plug connector **12f** and the second plug connector **12**, which will be described further below.

(41) The first plug connector **12f** includes the main body **20f**, the push member **22f**, the retainer lug **24f**, and the guide ribs **30f**. Configurations of the main body **20f**, the push member **22f**, the retainer lug **24**, and the guide ribs **30f** are almost identical to those of the first plug connector **12f** shown in FIGS. **1** and **3**, and the descriptions related to the configurations are not repeated. It should be noted that the first plug connector **12f** in this example does not include the coupler member **32**, in contrast to the first plug connector **12f** shown in FIGS. **1** and **3**.

(42) The second plug connector **12s** has a configuration almost identical to the first plug connector **12f**, and includes the main body **20s**, the push member **22s**, the retainer lug **24s**, and the guide ribs **30s**. However, the guide ribs **30s** are arranged at positions shifted toward the center in the second direction relative to the guide ribs **30f**. That is, a positional relationship between the guide ribs **30s** and the main body **20s** is different from a positional relationship between the guide ribs **30f** and the main body **20f**.

(43) The terminal base **15** shown in FIG. **6** has a first receptacle connector **14f** configured to receive the first plug connector **12f** and a second receptacle connector **14s** configured to receive the second plug connector **12s**. The first receptacle connector **14f** and the second receptacle connector **14s** have a very similar formation, yet have a difference in position between guide grooves **52f** and guide grooves **52s** that are configured to receive the guide ribs **30f** and **30s**, respectively.

(44) For this reason, when it is attempted to insert the first plug connector **12f** into the second receptacle connector **14s**, the guide ribs **30f** interfere with a part of the second receptacle connector **14s**. Similarly, when it is attempted to insert the second plug connector **12s** into the first receptacle connector **14f**, the guide ribs **30s** interfere with a part of the first receptacle connector **14f**. In other words, an insertion error where the plug connector **12f** or **12s** is inserted into an unassociated one of the receptacle connectors **14s** and **14f** can be reliably prevented by varying the positional relationship between the guide ribs **30f** and the main body **20f** from the positional relationship between the guide ribs **30s** and the main body **20s**.

(45) When inserting the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s**, the

operator binds a relative positional relationship between the plug connectors **12f** and **12s** by the jig **60**. The jig **60** is a substantially U-letter shaped member which is open at one of its second direction ends when viewed from above. In other words, the jig **60** is open to the outside at the one of the second direction ends and has a cutout space **64** penetrating through the jig **60** along the vertical direction. The cutout space **64** is defined to allow the electric wires **18f** and **18s** to pass therethrough.

(46) In addition, a pair of binding walls **65** are downwardly protruded from a lower end surface of the jig **60**. The pair of binding walls **65** are spaced from each other along the first direction, and a spacing distance in the first direction between the two binding walls **65** is substantially equal to or slightly greater than a dimension, in the first direction, of the main body **20**. In addition, each of the binding walls **65** has two binding grooves **66f** and **66s** which are spaced from each other in the second direction and configured to receive the push members **22f** and **22s**, respectively.

Accordingly, in the entire jig **60**, a total of four binding grooves **66f** and **66s** are formed.

(47) The binding grooves **66f** and **66s** respectively receive partial regions (i.e., the push members **22f** and **22s**) in the two plug connectors **12f** and **12s** being adjacently aligned along the second direction, to thereby function as a binding member for binding the relative positional relationship between the two plug connectors **12f** and **12s**. Dimensions, in the second direction, of the binding grooves **66f** and **66s** are defined to be approximately equal to or slightly smaller than dimensions in the second direction of the push members **22f** and **22s**. In this connection, as shown in FIGS. 5 and 9, walls of the binding grooves **66f** and **66s** located at second direction ends of each of the binding walls **65** are relatively thin and bendable like a leaf spring. When the push members **22f** and **22s** are respectively press fitted into the binding grooves **66f** and **66s**, the push members **22f** and **22s**, and thus the plug connectors **12f** and **12s**, are retained due to a frictional force or an elastic restoring force of the leaf spring by the jig **60**. Further, bottom surfaces of the binding grooves **66** function as contact surfaces **68** for pushing the pushing surfaces **28**.

(48) In operation to mate the connectors, the operator positions the main bodies **20f** and **20s** between the pair of the binding walls **65** and then forces the push members **22f** and **22s** to enter the binding grooves **66f** and **66s**. As a result, the pair of binding walls **65** inhibit movement of the plug connectors **12f** and **12s** in the first direction while the binding grooves **66f** and **66s** inhibit movement of the plug connectors **12f** and **12s** in the second direction. In addition, the plug connectors **12f** and **12s** are retained within the jig **60** by the frictional force occurring between the binding groove **66f**, **66s** and the push member **22f**, **22s** or the elastic restoring force of the leaf spring, without escaping downward from the jig **60**. As a result, the two plug connectors **12f** and **12s** are bound to a positional relationship in which the plug connectors **12f** and **12s** are adjacent to each other in the second direction.

(49) In this state, the operator places the plug connectors **12f** and **12s** in position relative to the associated receptacle connectors **14f** and **14s** and pushes the jig **60** from above. This causes the plug connectors **12f** and **12s** to respectively enter the associated receptacle connectors **14f** and **14s**. In this operation, the four pushing surfaces **28** are simultaneously pushed by the contact surfaces **68** in the jig **60**. Here, a top surface of the jig **60** has a sufficient space to be held by a hand or fingers, which allows the operator to planarly push and press the jig **60**. It is therefore possible to downwardly push the plug connectors **12f** and **12s** while preventing the jig **60**, and thus the plug connectors **12f** and **12s**, from getting inclined. As a result, there is no necessity to constantly adjust the position of the plug connectors **12f** and **12s** during operation to insert the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s**, and workability in operation to mate the connectors can be accordingly improved.

(50) It should be noted that the four binding grooves **66f** and **66s**, all of which have the same shape, might invite a fitting error which is caused by incorrectly fitting the push member **22s** into the binding groove **66f** being essentially intended to receive the push member **22f**. However, if an attempt to insert the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s** is made

in such an erroneously fitted state, the guide ribs **30f** and **30s** will interfere with a part of the receptacle connectors **14f** and **14s**, and accordingly prevent the attempt to insert the plug connectors **12f** and **12s** from being successfully accomplished. In this way, an insertion error of the plug connectors **12f** and **12s** can be reliably prevented according to this example.

(51) When the two plug connectors **12f** and **12s** enter the receptacle connectors **14f** and **14s** to the deepest position, the retainer lug **24** is locally engaged with the receptacle connectors **14f** and **14s**. After the engagement is established, the operator pulls up the jig **60** to remove the jig **60** from the plug connectors **12f** and **12s**. Here, because the jig **60** has a holding force which is sufficiently smaller than an engagement force exerted by the retainer lug **24**, it is possible to remove the jig **60** from the plug connectors **12f** and **14s** only by lifting the jig **60**. When the jig **60** is removed in this way, the connector mating work is complete.

(52) As can be clearly understood from the above explanation, the relative positional relationship between the two plug connectors **12f** and **12s** is bound by the jig **60**, and the thus-bound plug connectors **12f** and **12s** can be planarly pushed and pressed from the top of the jig **60**. As a result, the plug connectors **12f** and **12s** can be effectively prevented from getting inclined, which makes the work to insert the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s** quick and easy.

(53) The above-described configuration is presented by way of illustration, and may be modified in various ways other than features that each of the plug connectors **12f** and **12s** has the pair of push members **22f**, **22s** protruding along the first direction from both sides of the main body **20f**, **20s**, and that the two plug connectors **12f** and **12s** are mechanically coupled to each other or bound by the jig **60** in a position where the push members **22f** are displaced in the second direction from the push members **22s**, and are inserted into the associated receptacle connectors **14f** and **14s** while being mechanically coupled to each other or bound by the jig **60**.

(54) For example, the shapes and other features of the plug connectors **12f** and **12s** may be changed as appropriate. For example, the second plug connector **12s**, which does not include the retainer lug **24** in the example shown in FIG. 1, may have a retainer lug **24** on a surface located opposite the coupling member **38**. Further, in the example of FIG. 1, the way of coupling the two plug connectors **12f** and **12s** may be changed as appropriate. For example, the two plug connectors **12f** and **12s** may be coupled to each other by a magnetic force or a hook-and-loop fastener, by press fitting, or by other component. Also in the example shown in FIG. 5, the shapes of the plug connectors **12f** and **12s** and the jig **60** may be changed as appropriate.

(55) In the above description, the connector set **10** includes the two plug connectors **12f** and **12s**, but may be configured to include three or more plug connectors **12**. For example, the connector set **10** may include three plug connectors which can be mechanically coupled to each other. In addition, the number of electric wires **18f** or **18s** to be connected to each of the plug connectors **12f** or **12s**, and arrangement, types, and other features of the electric wires **18f** and **18s**, may be changed as appropriate. Therefore, the electric wires **18f**, **18s** may be an electric wire connected to any electronic device other than the temperature sensor.

REFERENCE SIGNS LIST

(56) **10** connector set; **12f** first plug connector; **12s** second plug connector; **14f** first receptacle connector; **14s** second receptacle connector; **15** terminal base; **16f**, **16s** housing; **18f**, **18s** electric wire; **20f**, **20s** main body; **22f**, **22s** push member; **24f**, **24s** retainer lug; **26f**, **26s** wire connection hole; **28f**, **28s** pushing surface; **30f**, **30s** guide rib; **32** coupler member; **34** coupling loop; **36** slide claw; **38** coupling member; **40** rail member; **42** coupling lug; **42a** hitching portion; **44** slide rail; **45** stopper surface; **52f**, **52s** guide groove; **60** jig; **64** cutout space; **65** binding wall; **66f**, **66s** binding groove; **68** contact surface.

Claims

1. A connector mating method for inserting a first plug connector into a first receptacle connector and a second plug connector into a second receptacle connector, to mate the first and second plug connectors to the first and second receptacle connectors, wherein each of the first plug connector and the second plug connector comprises a main body configured to retain a plurality of electric wires, and a pair of push members protruding from both sides of the main body along a first direction, wherein the first plug connector and the second plug connector are mechanically coupled to each other or bound by a jig to a relative positional relationship therebetween in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction, and inserted into the first receptacle connector and the second receptacle connector while being mechanically coupled to each other or being bound to the relative positional relationship by the jig, wherein the main body comprises a plurality of wire connection holes in which the plurality of electric wires are respectively inserted, the plurality of wire connection holes and the pair of push members being aligned with each other along the first direction, wherein the first receptacle connector and the second receptacle connector are recesses that are completely separated from each other, with a partition wall interposed between the first receptacle connector and the second receptacle connector, and wherein, upon the first plug connector and the second plug connector being mechanically coupled to each other or bound by the jig, a recess for receiving the partition wall is formed between the first plug connector and the second plug connector.
2. The connector mating method according to claim 1, wherein the first plug connector and the second plug connector are mechanically coupled to each other or bound to the relative positional relationship by the jig in a position where the first and second plug connectors are adjacent in the second direction.
3. The connector mating method according to claim 1, wherein each of the first and second plug connectors further comprises a guide rib extending along an insertion direction, the guide rib being configured to be received in a guide groove defined in an associated one of the first and second receptacle connectors, wherein a relative position between the guide rib in the first plug connector and the main body is varied from a relative position between the guide rib in the second plug connector and the main body, and wherein when one of the plug connectors tries to enter an unassociated one of the first and second receptacle connectors, the guide rib interferes with a part of the unassociated one of the first and second receptacle connectors to thereby block further entry of the one of the plug connectors.
4. A connector set, comprising: a first plug connector configured to be inserted into a first receptacle connector; and a second plug connector configured to be inserted into a second receptacle connector, wherein each of the first plug connector and the second plug connector comprises a main body configured to retain a plurality of electric wires, and a pair of push members protruding from both sides of the main body along a first direction, wherein the first plug connector and the second plug connector are mechanically couplable to each other in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction, wherein the main body comprises a plurality of wire connection holes in which the plurality of electric wires are respectively inserted, the plurality of wire connection holes and the pair of push members being aligned with each other along the first direction, wherein the first receptacle connector and the second receptacle connector are recesses that are completely separated from each other, with a partition wall interposed between the first receptacle connector and the second receptacle connector, and wherein, upon the first plug connector and the second plug connector being mechanically coupled to each other, a recess for receiving the partition wall is formed between the first plug connector and the second plug connector.
5. A connector set, comprising: a first plug connector configured to be inserted into a first

receptacle connector; a second plug connector configured to be inserted into a second receptacle connector; and a jig, wherein each of the first plug connector and the second plug connector comprises a main body configured to retain a plurality of electric wires, and a pair of push members protruding from both sides of the main body along a first direction, wherein the jig comprises a binding member configured to partially receive both the first plug connector and the second plug connector for binding a relative positional relationship between the first plug connector and the second plug connector in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction, wherein the binding member includes a pair of binding grooves that receive two of the push members and a single binding wall defining a respective edge of the pair of binding grooves in the second direction, wherein the first receptacle connector and the second receptacle connector are recesses that are completely separated from each other, with a partition wall interposed between the first receptacle connector and the second receptacle connector, and wherein, upon the first plug connector and the second plug connector being bound by the jig, a recess for receiving the partition wall is formed between the first plug connector and the second plug connector.
